An attractor account of numerosity readout

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Abstract

Numerical perception is often described as the product of two systems: a near-exact mechanism for small sets and an approximate one for larger sets, whose variability follows Weber's law. Gallistel and Gelman proposed instead that both behaviours could arise from a single representation of discrete numerosities whose precision degrades progressively with magnitude. Here we give this idea a mechanistic form, where a noisy estimate of numerosity, already extracted by upstream perceptual processing, is turned into a discrete numerical response. Integers are represented as memories along a one-dimensional number line, jointly creating a Hopfield-inspired attractor landscape. This estimate provides the initial state, and recurrent dynamics clean it up by pulling it towards the stored integer representations. Because the memory components become broader and overlap more strongly with magnitude, the same landscape produces near-categorical responses for small numerosities and increasingly coarse, Weber-like estimates for larger ones.

We tested these predictions by reanalysing 52,995 individual responses from an openly available visual-enumeration dataset. The attractor model outperformed matched static and constant-width controls, a purely driven accumulator, and Weber-style and bounded-optimal alternatives. Response time followed naturally from the number of clean-up steps, with a finite stopping budget accounting for the large-number plateau. The fitted response distributions approached a stable Weber fraction at larger numerosities, and participants with sharper low-number landscapes showed lower Weber noise on a non-overlapping set of large-number trials. Together, these results suggest that magnitude-dependent landscape geometry and recurrent clean-up make separable contributions to a shared numerical readout.

Keywords

numerical cognition, subitizing, Hopfield network, attractor dynamics, approximate number system

Introduction

Humans, like many non-human animals, possess a non-symbolic ability to extract numerical information from the environment and represent the numerosity of sets, a capacity commonly termed the number sense (Dehaene 2011; Nieder 2016). This ability is typically expressed in two characteristic behavioural regimes: small sets, typically up to three or four items, are perceived rapidly and with high accuracy, a phenomenon known as subitizing (Kaufman et al. 1949; Mandler and Shebo 1982). Beyond this range, responses become slower and less accurate, response variability grows with set size, and discrimination depends primarily on the ratio between numerosities (Feigenson et al. 2004; Dehaene 2011). The standard two-systems account attributes these regimes to distinct mechanisms: a capacity-limited Object Tracking System, which individuates a few objects in parallel, and an Approximate Number System, which represents larger numerosities as noisy analogue magnitudes (Feigenson et al. 2002; Piazza et al. 2011).

An alternative account of the number sense treats both regimes as consequences of a single magnitude code for countable quantities. Gallistel and Gelman proposed that a non-verbal counting process generates analogue magnitudes through successive increments, thereby giving discrete quantities an ordered representation on a linear scale (Gallistel and Gelman 1992, 2000). These representations exhibit scalar variability: their standard deviation grows in proportion to magnitude, so neighbouring magnitude distributions overlap increasingly at larger values. Evidence from non-verbal production and rapid sequential-estimation tasks supports this scaling both within and beyond the usual subitizing range (Whalen et al. 1999; Cordes et al. 2001). Increasing overlap could therefore allow one code to support highly accurate small-number judgements while yielding progressively graded, ratio-dependent errors for larger numerosities.

This account describes a general cognitive model of non-verbal number representation, but it does not specify the underlying mechanisms. Their theoretical model alone does not specify how a noisy magnitude is stabilized and mapped back onto an integer response, or whether the exceptional performance observed for the smallest sets is generated by the same process. Gallistel and Gelman themselves left the status of numerosities below five as an open question (Gallistel and Gelman 2000). Their proposal therefore provides a representational starting point for

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a single-code account, while leaving the mechanism of discrete numerical readout unresolved.

Computational work has developed several single-system explanations of the two behavioural regimes. The central comparison for the present study is the pair of capacity-limited accounts developed by Cheyette, Piantadosi, and colleagues. At a normative level, Cheyette and Piantadosi (2020) ask how a system with finite information-processing capacity should encode numerosities under a power-law prior. The resulting bounded-optimal code allocates greater precision to small values and progressively coarser precision to larger ones, producing near-exact small-number performance and Weber-like variability without a change of numerical system. At a more visually explicit level, Cheyette et al. (2024) locate the relevant bottleneck in limits on visuospatial memory, which can generate the characteristic psychophysics of numerosity from visual arrays. Their within-participant analysis found that information-capacity estimates transfer between non-numerical change localisation and numerosity estimation, providing direct evidence for a shared visuospatial constraint. An earlier, representationally distinct model by Zorzi and Butterworth (1999) established a related point: numerical size and distance effects can emerge from a linear cumulative code coupled to nonlinear response competition, without assuming logarithmic compression or scalar representational noise. More recently, Malo et al. (2023) showed that a small quantum-spin model can encode numerosity in the spectrum of its collective dynamics, providing a distinct dynamical route to a unified numerical representation. Other models have derived numerical sensitivity, range structure, adaptation, and switch-cost phenomena from recurrent visual dynamics (Sengupta et al. 2014, 2017; Verma and Sengupta 2023), or learned image-computable representations (Stoianov and Zorzi 2012; Testolin et al. 2020). Together, these accounts show that the two behavioural regimes need not imply two numerical systems, while leaving open whether their common source lies in capacity-limited sensory encoding or in the structure and dynamics of a shared numerical representation.

These visual accounts treat numerosity as a product of modality-specific sensory processing, whether as a consequence of limited visuospatial capacity or as a quantity extracted from the image. Number, however, behaves as an abstract quantity that is not bound to any one sense. It transfers across modalities, so that adapting to a stream of sounds or touches shifts the apparent numerosity of a visual array (Arrighi et al. 2014), its acuity tracks numerical experience, sharpening with formal schooling in otherwise comparable adults (Piazza et al. 2013; Nys et al. 2013), and it depends causally on parietal cortex, where stimulating the intraparietal sulcus disrupts numerosity while sparing lower-level vision (Cappelletti et al. 2007; Dormal et al. 2012) and focal damage yields acalculias that extend even to enumerating touches (Ashkenazi et al. 2008; Cohen et al. 2018). This picture is not uniform: nonsymbolic number stays robust with little formal schooling once memory and symbolic demands are minimized (Frank et al. 2008; Zebian and Ansari 2012; Pica et al. 2004), so this dichotomy is still under debate. This debate motivates the exploration of an architecture in which the classical numerical effects emerge from a purely numerical representation that can be fed by modality-specific sensory processing.

We thus propose a one-dimensional, Hopfield-inspired energy landscape as this shared numerical stage (Hopfield 1982; Ramsauer et al. 2021). Each integer contributes a stored memory component, and these components jointly generate a sequence of attractor basins that becomes broader and shallower with magnitude (Fig. 1a). We motivate this landscape geometry by the cross-linguistic power-law decline in number-word frequency (Dehaene and Mehler 1992), so that more frequently encountered numbers form narrow, deep basins, whereas less frequent numbers form broad, shallow ones. A one-dimensional estimate y_0 of numerosity, supplied by modality-specific sensory processing, initializes the landscape, followed by recurrent clean-up that stabilizes this evidence through pattern completion (Fig. 1b). The attractor therefore models numerical readout: the transformation from noisy numerical evidence to an integer response, downstream of the perceptual extraction of numerosity from the stimulus. The same dynamics link response time to the number of clean-up steps (Fig. 1c).

On this view, a single numerical landscape supports two forms of readout. Where neighbouring integer attractors are sufficiently distinct, pattern completion stabilizes noisy evidence around a categorical value, yielding rapid and highly accurate responses. As magnitude increases, adjacent basins broaden, become shallower, and increasingly overlap, so categorical resolution is progressively lost; the same landscape then produces coarser, ratio-dependent estimates. The transition from resolved to approximate responding therefore emerges from the geometry of one representation. The model thus gives mechanistic form to the single-code architecture proposed by Gallistel and Gelman, while adding an explicit representational mechanism: integer-centred attractors embedded in a continuous, noisy state space (Gallistel and Gelman 1992, 2000).

We test the model by reanalysing 52,995 individual responses from the openly available brief-exposure dataset of Cheyette and Piantadosi (2020). It captures the progression from near-exact small-number reports to underestimation and Weber-like variability at larger numerosities. Model comparisons show that both magnitude-dependent broadening and recurrent clean-up contribute to this fit, while bounded convergence time accounts for the main response-time profile. Finally, across participants, sharper fitted landscapes are associated with lower Weber noise, consistent with a common representational geometry constraining both near-categorical and approximate numerical behaviour.

Methods

We consider the problem of how our model maps a noisy perceptual input onto a discrete numerical response. We posit an energy landscape defined over a continuous axis x, with attractor wells associated with integer positions (Fig. 1a). In the visual task analysed here, the initial state y_0 summarizes evidence extracted from a dot array; formally, the same interface could receive scalar evidence from another sensory modality. Starting from $x = y_0$, the system evolves by gradient descent on the energy landscape

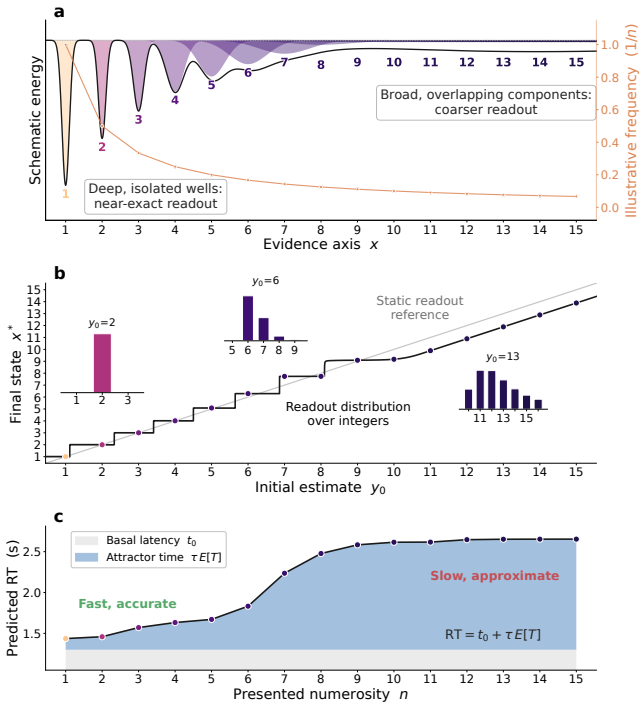


Figure 1. Schematic and fitted components of the attractor model. a, Schematic illustration of the shared fitted landscape. Gaussian-component widths follow the fitted power law, while widths and depths are floored for visibility; the orange $1/n$ curve illustrates the power-law decline in number-word frequency that motivates widening and is not an empirical frequency series. Small numbers therefore appear as deep, isolated components and larger numbers as broad, overlapping components. Figure 2c reports the width index n_{width}. **b**, Fitted attractor clean-up map from the initial estimate y_0 to the bounded final state $x^*(y_0)$. The light-grey diagonal is the matched static readout ($x^* = y_0$); the black curve is the transformation produced by clean-up under the fitted dynamics. Insets show the resulting distributions over integer responses, $p(r \mid x^*)$, for $y_0 = 2, 6, 13$. **c**, The selected response-time link, $\mathrm{RT} = t_0 + \tau\, E[T]$, evaluated with the 160-ms sensory parameters and decomposed into basal latency and convergence time. An extended integer support (1–30) is used in this panel to avoid edge truncation near the upper displayed numerosities.

until x settles into a stable well, implementing Hopfield-style pattern completion (Hopfield 1982). The resulting clean-up map $y_0 \mapsto x^*$ pulls nearby estimates towards supported integer basins while leaving poorly resolved regions closer to the identity (Fig. 1b). A probability distribution over integers is then read out from the final resting state x^*. The settling dynamics also determine how long the trial takes.

In our landscape, each integer n acts as a stored memory represented by a normalized Gaussian component centred at n. Its scale σ_n controls both the breadth and the effective depth of the resulting basin: increasing σ_n broadens the component and, through the normalization term $-\log \sigma_n$, lowers its peak contribution to the log-sum-exp landscape. We use the cross-linguistic power-law decline in number-word frequency (Dehaene and Mehler 1992) to motivate this coupled scaling, so frequent small numbers form narrow, deep basins and rarer large numbers form broad, shallow ones. Accordingly, we assume that the Gaussian scale grows as a power of magnitude,

$$\sigma_n = \sigma_c\, n^p, \tag{1}$$

where σ_c sets the overall scale and the exponent p controls how fast resolution degrades with magnitude; both are free parameters fitted to the responses. Each memory component contributes a log-density

$$g_n(x) = -\log \sigma_n - \frac{(x-n)^2}{2\,\sigma_n^2}. \tag{2}$$

Using the log-sum-exp energy form associated with modern Hopfield networks (Ramsauer et al. 2021), the attraction potential is the log-sum-exp over all wells, $V(x) = -\log \sum_n \exp\big(g_n(x)\big)$.

A sensory estimate $y_0 \sim \mathcal{N}\big(n,\ s_{00}^2 + s_0\, n^{\lambda_0}\big)$, whose noise grows with numerosity ($s_{00} = 0.05$ fixed; s_0 and λ_0 fitted), starts the descent on integer wells at $m = 1, \ldots, 19$. From $x = y_0$ the state descends the potential, $x_{t+1} = x_t - \eta\, \partial V/\partial x$ with fixed $\eta = 1$ and a global step clamp (full numerical settings in Supplementary Material 1). Because V is a log-sum-exp over the stored integers, each step moves the state towards a softmax combination of the wells, weighted by their precisions, so every stored integer contributes to the final position.

The descent stops when the update falls below a fixed tolerance or reverses sign (oscillation). The reversing or sub-threshold update is not applied: x^* is the current state, and T is the recorded stopping-iteration count, including the iteration on which stopping is detected. On the almost flat, large-numerosity landscape, some trajectories do not meet this numerical settling criterion within an iteration budget compatible with the observed response times. We therefore introduce a hypothesized give-up mechanism that stops the clean-up process after G steps, giving $T = \min(T^*, G)$, where T^* is the unbounded settling time. We fix G at the profile-likelihood optimum, $G = 120$ (Supplementary Material 1). A distribution over integer responses is then read out, $p(r \mid x^*) = \mathrm{softmax}_r\big(g_r(x^*)\big)$, from which we take the mean estimate and the most probable integer.

The model gives a natural interpretation of response time through the time required for the clean-up mechanism to settle on a well:

$$\mathrm{RT} = t_0 + \tau\, \mathbb{E}[T], \tag{3}$$

where T is the recorded stopping-iteration count, bounded by the global budget G, and the expectation is taken over sensory noise. Observed response times are modelled with a log-normal likelihood for which this value is the median and geometric mean, rather than the arithmetic mean.

To test whether static readout uncertainty contributed beyond the attractor dynamics, we compared this convergence-time link with an entropy-only alternative, $t_0 + \kappa \mathbb{E}[H]$, and with the hybrid $t_0 + \tau \mathbb{E}[T] + \kappa \mathbb{E}[H]$. All mapping coefficients were shared across exposure durations, whereas the residual RT variance was duration-specific. The basal latency $t_0 = 1.30$ s is task-fixed: recorded RT starts at stimulus offset, so t_0 absorbs mask, typing, and motor time rather than exposure (Supplementary Material 1).

We fit the model to the individual responses of Cheyette and Piantadosi (2020), in which adult participants briefly viewed dot arrays of 1–15 items and typed an estimate, at five

exposure durations (40, 80, 160, 320, 640 ms). The source paper analysed four between-experiment arms at $N = 100$ each after excluding the ten highest-error participants per arm; the open DURATION file used here pools the three duration-control arms without that exclusion and is within-subject across exposures (379 sessions; 360 completed all five durations). We evaluate the categorical likelihood of the 52,995 individual integer reports. For each numerosity the response probability marginalizes the readout over the sensory noise,

$$p(r \mid n) = \int p\big(r \mid x^*(y_0)\big)\, \phi\big(y_0; n, \sigma(n)\big)\, dy_0, \qquad (4)$$

with $\sigma(n) = \sqrt{s_{00}^2 + s_0\, n^{\lambda_0}}$, computed by deterministic grid quadrature (Supplementary Material 1). Typed responses run well beyond the well support, so we add a uniform lapse channel, $p_{\text{final}}(r \mid n) = (1-\ell)\, p(r \mid n) + \ell \,/\, R_{\max}$, with ℓ fitted per duration and counted in all information criteria. In the joint fit the landscape (σ_c, p) and give-up budget G are shared across durations, while (s_0, λ_0, ℓ) are duration-specific ($k = 18$; Supplementary Material 2), fitted by differential evolution (Storn and Price 1997) and Nelder–Mead polishing. With the landscape held fixed, the selected RT channel adds one shared convergence-time coefficient τ and a per-duration residual σ_{rt}. Full numerical settings and fitted values are in Supplementary Material 1.

We fitted the model with a joint fit in which a single power-law landscape (σ_c and p) and the give-up budget G were shared across all five exposure durations, while the sensory channel and lapse were allowed to vary by duration. Models were compared under the same trial-level likelihood against a graded set of alternatives. Matched static readouts isolate the contribution of clean-up for both width laws, whereas constant-width controls fix $p = 0$ to isolate the contribution of magnitude-dependent broadening (as opposed to the effect of increasing noise variance). A Weber/ANS code and the bounded-optimal model of Cheyette and Piantadosi (2020) benchmark the account as a whole. Parameter sharing, total model flexibility, and the corresponding AIC and BIC values are reported in Supplementary Table 2.

To test whether the attractor could itself implement numerical accumulation, we fitted a purely driven variant with no completed scalar estimate y_0. During exposure, registered items generated a running count $K(t)$, which drove the state while the attractor force acted,

$$\tau_x \dot{x} = -\frac{\partial V(x)}{\partial x} + \beta\big(K(t) - x\big).$$

At stimulus offset the drive was removed, and the response was read from the resulting state. This variant tests whether serial evidence accumulation and associative clean-up can be implemented by the same dynamics.

We summarize the fitted width law with the width index n_{width}, defined by $\sigma_n = 1$, where the fitted Gaussian-component width equals the unit spacing between integers. With $\sigma_n = \sigma_c n^p$, this gives $n_{\text{width}} = \sigma_c^{-1/p}$. The definition is equivalent to the heuristic $d' = \text{spacing}/\sigma_n = 1$ (Macmillan and Creelman 2004) and summarizes the combined effect of the fitted scale and exponent.

For the individual-differences analysis, all five exposure durations informed two non-overlapping response channels: $n \le 8$ estimated participant effects on lower-numerosity n_{width}, and $n \ge 9$ estimated participant effects on Weber noise. Population means were duration-specific, while participant effects on $\log n_{\text{width}}$, $\log w$, and logit lapse were shared across durations and jointly Gaussian. We report the latent n_{width}–w correlation conditional on participant lapse (Supplementary Material 4).

Results

Among the models considered, the power-law attractor provided the best account of the trial-level responses (Fig. 2a; Supplementary Material 2). The constant-width attractor trailed by $\Delta\text{AIC} \approx 3488$, the Weber/ANS code by $\Delta\text{AIC} \approx 838$, and the bounded-optimal model of Cheyette and Piantadosi (2020) by $\Delta\text{AIC} \approx 11{,}453$. The attractor dynamics require one extra "budget" parameter, but, if we switch them off so that we have a static readout of the same wells, the model fits worse by $\Delta\text{AIC} \approx 722$. The results thus indicate that the dynamics are an important element of the model, and motivate the introduction of this "give-up" mechanism represented by the budget parameter G. Finally, the purely driven accumulator fitted substantially worse than the completed-estimate attractor ($\Delta\text{AIC} = 7954$; Fig. 2a). In this purely driven model, as the running count increased, the state was repeatedly drawn towards intermediate integer representations and often had not reached the presented numerosity by stimulus offset (Supplementary Fig. S1).

Because clean-up is a dynamical process, its convergence time provides a direct measure of response time. Adding the entropy term did not improve the fit: the hybrid was 0.24 AIC points worse than convergence time alone, so we retained the simpler model. An entropy-only link was 1579.5 AIC points worse than convergence time (Fig. 2b). Therefore, residual readout entropy did not add supported predictive value beyond convergence time. Nevertheless, clean-up on the flat large-numerosity landscape can fail to meet the settling criterion within a behaviourally plausible time budget; allowing every trajectory to reach its eventual limit would then imply implausibly long latencies. The give-up mechanism caps clean-up after G steps, so the predicted response time plateaus with the data once the budget is reached (Fig. 3c, vertical line).

The fitted landscape has wells with width growing with magnitude, so they are sharp and well separated for small numbers and broad and overlapping for large ones. A joint parameter-recovery analysis accurately recovers both σ_c and p, although their estimation errors remain negatively correlated because both determine the same width law (Supplementary Material 3). We therefore summarize their combined effect with the operational index $n_{\text{width}} = \sigma_c^{-1/p}$. In the joint fit, the shared width law gives $n_{\text{width}} = 8.5$, which falls near the onset of the response-time plateau (Fig. 1c).

To test whether the same representational geometry constrains lower- and higher-numerosity behaviour across individuals, we related landscape resolution to Weber noise (Fig. 2c). Higher n_{width} remained associated with lower w after conditioning on the participant lapse effect ($\hat{\rho}_{nw\cdot\ell} = -.63$, $N = 360$; likelihood-ratio $\chi^2(1) = 105.0$, $p = 1.2 \times 10^{-24}$). The fitted response distributions also reproduced

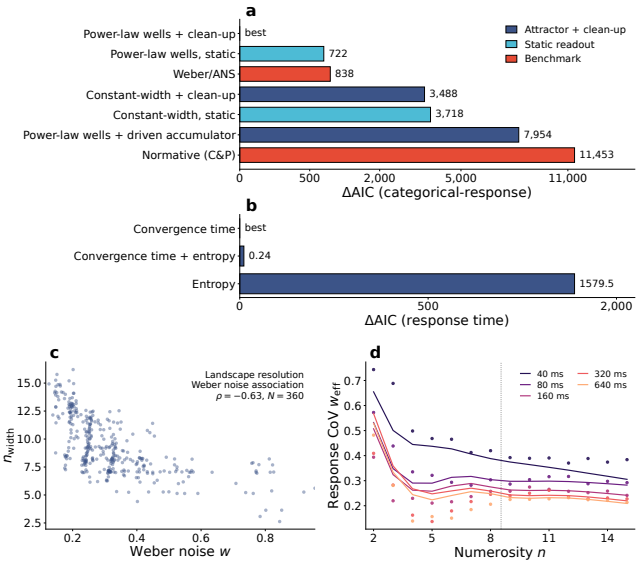


Figure 2. Model comparison and the link between landscape width and estimation acuity. a, Categorical-response channel: ΔAIC of each model relative to the winning power-law attractor with clean-up, over the ∼53,000 individual responses. Bars are coloured by model role: attractor with clean-up, matched static readout of the same wells, or external benchmark. Turning clean-up off, while retaining the same landscape and sensory parameterization but requiring no stopping-budget worsens the fit, and constant-width wells ($p = 0$, $\sigma_n = \sigma_c$) worsen it further. The purely driven attractor ("Power-law wells + driven accumulator", see Methods), whose input is the still-accumulating count rather than a completed estimate, fits far worse, because a partial count captures the wrong integer (Supplementary Material 2). Weber/ANS and the bounded-optimal model of Cheyette and Piantadosi (2020) ("Normative (C&P)") are shown for reference. **b**, Response-time channel: ΔAIC of the three RT links on the winning landscape, with one mapping shared across durations. Convergence time alone and the hybrid link perform almost identically, whereas entropy alone performs substantially worse. **c**, Individual differences in the 360 sessions completing all five durations. Non-overlapping numerosity ranges informed landscape resolution ($n \leq 8$) and Weber noise ($n \geq 9$). Points show duration-averaged empirical-Bayes estimates; the inset reports their latent correlation conditional on the participant lapse effect. **d**, Effective Weber fraction: the coefficient of variation of the fitted response distribution per numerosity and exposure duration. Solid lines are the model, dots the data, and the dotted vertical marks the shared n_{width}; the coefficient of variation approaches a roughly constant plateau at larger numerosities.

the Weber-like variability of larger-numerosity estimates, where their coefficient of variation approached a roughly constant plateau with increasing numerosity (Fig. 2d). Over $n \geq n_{\text{width}}$, the fitted exponent was shallow in both model and data ($\gamma = -.20$ and $-.10$, respectively), indicating approximately scalar variability across the larger-numerosity range.

The sensory channel is only phenomenological: s_0 sets the overall noise level of the initial estimate y_0, whereas λ_0 determines how that noise grows with numerosity. These parameters vary across exposure durations so that changes in input quality are not forced onto the shared landscape, and we do not assign them a further cognitive interpretation.

With these ingredients, the model captures all three behavioural measures (Fig. 3). Pooled across durations, it gives $R^2 = 0.95$ for the mean estimate, 0.94 for exact accuracy, and 0.82 for response time. The response-time analysis provides a constrained cross-channel test: the landscape is first fitted to the categorical responses and then held fixed, so that only the mapping from convergence time to observed RT is estimated from the response times.

The main deviations are concentrated in the 40-ms condition, particularly in the mean estimates and response times. At such brief exposures, even small numerosities show the graded variability normally associated with larger sets. Cheyette and Piantadosi (2020) similarly found that performance at the shortest exposures reflects a severely information-limited regime, and noted that human behaviour changes more sharply with exposure time than predicted by a simple linear accumulation process. The remaining mismatch may therefore reflect early temporal properties of visual encoding that are not captured by our phenomenological sensory channel.

A second residual is a shoulder predicted around $n = 6$–9 in exact accuracy and $n = 5$–8 in response time. This structure is generated by the attractor dynamics but is not clearly present in participants' behaviour (Fig. 3b,c; Supplementary Fig. S3). We return to its implications for the homogeneity and smoothness of the fitted landscape in the Discussion.

Discussion

We have described a model of the inference of numerosity from noisy scalar evidence as pattern completion on an attractor landscape: the integers act like memory components, made of attractor wells whose widths grow with magnitude, and an uncertain estimate relaxes towards the nearest resolvable integer. Fitted to trial-level visual-enumeration responses, a single landscape shared across exposure durations captures mean estimation, exact accuracy, and the principal response-time profile, while outperforming matched static, constant-width, Weber-style, bounded-optimal, and purely driven alternatives. Within the numerical readout, more categorical and more approximate reports therefore become different operating regions of one representation rather than outputs of two numerical codes joined at an unexplained boundary.

The magnitude-dependent width profile, which the trial-level comparison favours over constant-width wells, explains how this common representation changes its effective resolution. At low numerosities, neighbouring components are narrow and sufficiently isolated for clean-up to concentrate probability on a single integer. As magnitude increases, component overlap grows, exact-response probability declines, and the resulting response distributions approach the roughly constant coefficient of variation characteristic of nonsymbolic estimation (Whalen et al. 1999; Cordes et al. 2001; Testolin and McClelland 2021). The main residual of this account is a shoulder around $n = 5$–9 generated by clean-up on the integer lattice but less evident in the behavioural data. The empirical transition may therefore be smoother than in the fitted homogeneous landscape, potentially because

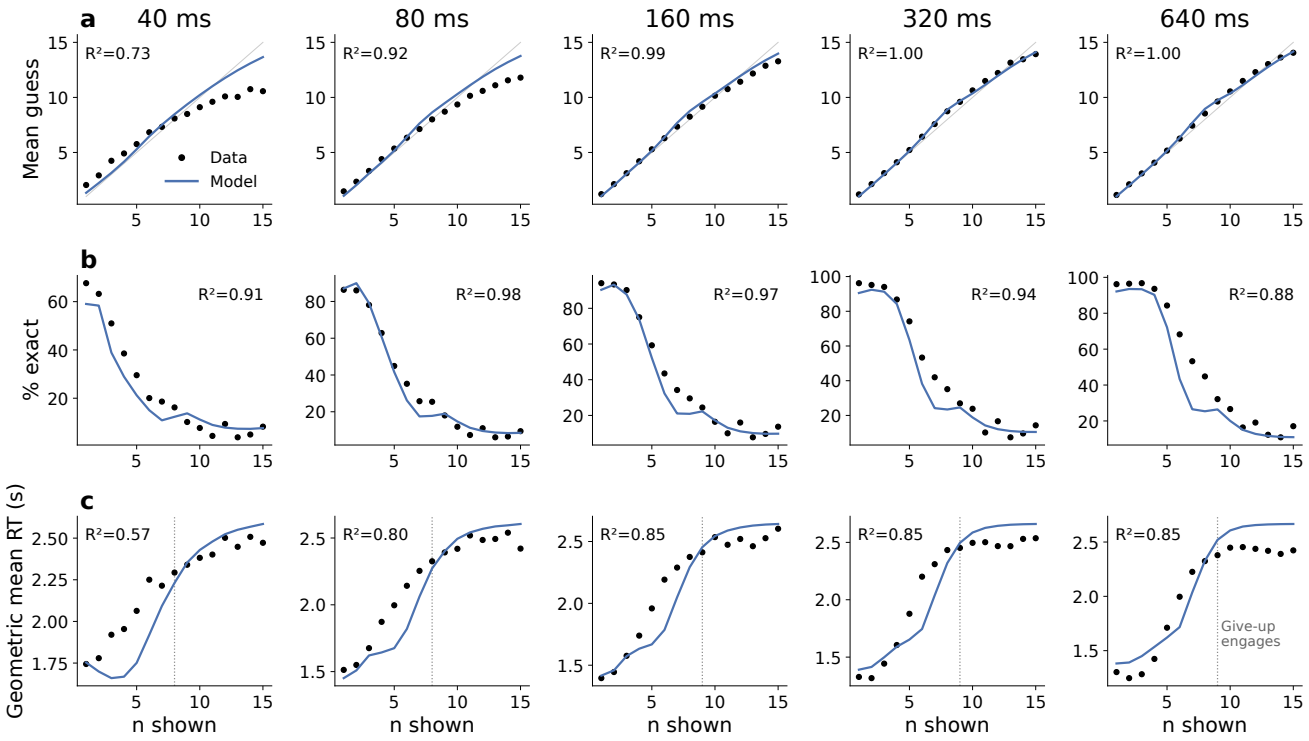


Figure 3. Fit of the joint model to the Cheyette and Piantadosi (2020) data across the three channels. Columns show presentation times from 40 to 640 ms; rows show the mean estimate, exact accuracy $P(r = n)$, and geometric-mean response time. Dots are empirical data; lines are the model, with the landscape fitted to the individual responses. Response time is evaluated in a constrained cross-channel fit: the landscape is held fixed at its response-fit values, and only the RT-link parameters are fitted to the individual trial times. The small shoulder in predicted exact accuracy occurs near the width-based proxy n_{width} and disappears when clean-up is switched off (Supplementary Fig. S3). Dotted vertical lines in the response-time row mark the smallest numerosity at which the fitted expected settling time exceeds the shared budget G, indicating where the give-up cap begins to bind. R^2 values are given per panel.

participants differ in landscape geometry or in when they stop deliberating. The fitted power-law widening is defined on a linear coordinate, but the present data do not uniquely identify this geometry: an appropriately compressed coordinate with a different width rule could provide a related description.

The matched static control shows that the attractor dynamics contribute beyond the width profile itself. In the static model, sensory variability passes directly into the readout, whereas clean-up instead maps nearby estimates that fall within the same basin towards a common final state before the response is generated. Where integer components are well separated, this compresses the variability of the input: a broader range of noisy estimates yields a readout concentrated around the same integer, increasing exact-response probability and narrowing the distribution of reports. As neighbouring components broaden and overlap, the basins become less selective and this stabilizing effect weakens. The Hopfield analogy nevertheless has limits: the purely driven accumulator fails to capture the data, implying that accumulation and numerosity retrieval are separate processes. This result is understandable because a partial count is not a degraded representation of the final cardinality: each prefix is itself a valid numerosity and remains compatible with every larger total until the set or sequence is complete. The attractor therefore cannot recover the final number from the running count alone. In the present architecture, modality-specific processes first construct an estimate of a delimited set or completed sequence, after which the shared landscape stabilizes that estimate.

The chronometric result provides a second consequence of the same dynamics. Under a single latent-to-time mapping shared across exposure durations, convergence time accounts for response time as well as the hybrid convergence-plus-entropy link and substantially better than residual entropy alone. The selected model therefore ties latency directly to the amount of attractor clean-up required, similarly to Zorzi and Butterworth (1999). In the flat large-numerosity region, however, full convergence can require more iterations than a behaviourally plausible response process would allow. We therefore introduce a simple "give-up" mechanism that truncates clean-up after a maximum of G steps. This converts the otherwise continuing increase in settling time into the observed response-time plateau. The plateau thus reflects the interaction between slowing attractor dynamics and a finite stopping policy: the present model captures the behavioural effect of that policy, while leaving its cognitive implementation open.

The width profile provides a potential representational locus for individual and developmental differences in numerical acuity. In the present data, participants with a sharper fitted landscape at lower numerosities also show less Weber noise on a non-overlapping set of larger-numerosity trials. This association is consistent with a common geometry constraining both categorical and approximate behaviour, although the evidence remains correlational.

The memory-like integer components, and this link with individual Weber noise, imply that one possible origin of differences in numerical acuity (and thus mathematics achievement (Halberda et al. 2008)) is numerical experience: if memory precision adapts to environmental statistics, frequently encountered numerosities should be represented more sharply than rarer ones (Anderson and Schooler 1991; Verguts and Fias 2004). This experience need not be symbolic or scholastic: small numerosities dominate everyday environments across modalities and formats, so the relevant statistics are available to unschooled humans and to other animals, in which numerosity tuning arises without numerical training (Viswanathan and Nieder 2013). On this view, development and numerical training should then progressively modify the width profile and improve acuity. This prediction concerns experience with *numerosities*: an intervention that sharpens the numerical landscape should produce different transfer from one that improves broader visuospatial perception (Butterworth and Laurillard 2017).

At the same time, the memory analogy has an important constraint. Integer memories are not independent patterns that can be strengthened arbitrarily: they occupy ordered positions in a one-dimensional Hopfield landscape, so changing one representation may alter neighbouring wells, basin boundaries, and the global resolution profile. This structure makes targeted training—and potentially the remediation of some number-sense impairments—an attractive possibility, but it also leaves open how the landscape could be sharpened as a whole without distorting its numerical organization.

The omission of a sensory front end is deliberate because the present study was designed to isolate the computations performed after scalar numerical evidence has become available. This separation matters because learned visual systems can already extract numerosity-selective features from unlabelled pixel-level input: Stoianov and Zorzi (2012) showed that such representations can emerge through unsupervised visual learning, and subsequent image-computable models reproduce effects of continuous visual magnitudes and dyscalculic profiles (Testolin et al. 2020; Dolfi et al. 2024). Adding a visual extractor to the present first test would therefore make it difficult to determine whether predictive success arose from sensory feature construction or from numerical pattern completion. Isolating the attractor avoids that ambiguity, but also defines the model's scope: it does not explain segmentation, continuous image statistics, spatial configuration, attentional load, or other perceptual contributions to classical subitizing effects. Evidence that attention selectively supports performance in the subitizing range, from dual-task costs (Burr et al. 2010) to the loss of the small-number advantage in speeded saccadic choices (Castaldi et al. 2020), fits this division of labour: on the present account, such manipulations degrade the evidence delivered to the landscape, much as brief exposure does in the data modelled here. A complete perceptual account should couple the landscape to an image-computable or recurrent visual front end and test whether attractor clean-up predicts structure beyond the front end's own readout. Parallel auditory and tactile implementations would then provide a direct test of whether modality-specific sensory channels converge on a common numerical landscape. Providing a sensory input to the current model is a natural direction for future work.

Numerical readout, in this framework, is pattern completion on a landscape of memories of the integer numbers. Categorical and approximate reports become two readings of one landscape rather than two codes joined at an unexplained seam. The bounded convergence time of the clean-up dynamics captures the main response-time profile, although the cognitive implementation of the stopping policy remains open. The result is a compact, mechanistic set of hypotheses about how the mind represents number.

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Author Contributions

Eudald Correig-Fraga: Conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing - original draft, and writing - review and editing.

Ethical Considerations

This article reports computational modelling and secondary analysis of a previously published, publicly available dataset. No new human participants were recruited, and no new human data were collected for this study. The original study was approved by the University of California, Berkeley Institutional Review Board and complied with relevant ethical regulations, as reported by Cheyette and Piantadosi (2020).

Consent to Participate

The source publication reports that all participants provided informed consent before taking part. No new participants were recruited for this secondary analysis.

Consent for Publication

Not applicable. This study does not contain identifiable individual participant information.

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Declaration of conflicting interests

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Data Availability

Code is available at https://github.com/eudaldc/hopfield_numbers. The source datasets are available at https://osf.io/svcy5/.

Supplemental material

Supplementary material is available with this article.

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